EX.No.-2.2 – CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

DATE:14.10.2026

AIM:

To create a 3D solid model from the given isometric drawing using Autodesk Fusion 360 software.

REQUIREMENTS:

1. Autodesk Fusion 360
2. Windows 10 operating system
3. 16 GB RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download and 500 Kbps or faster upload speed

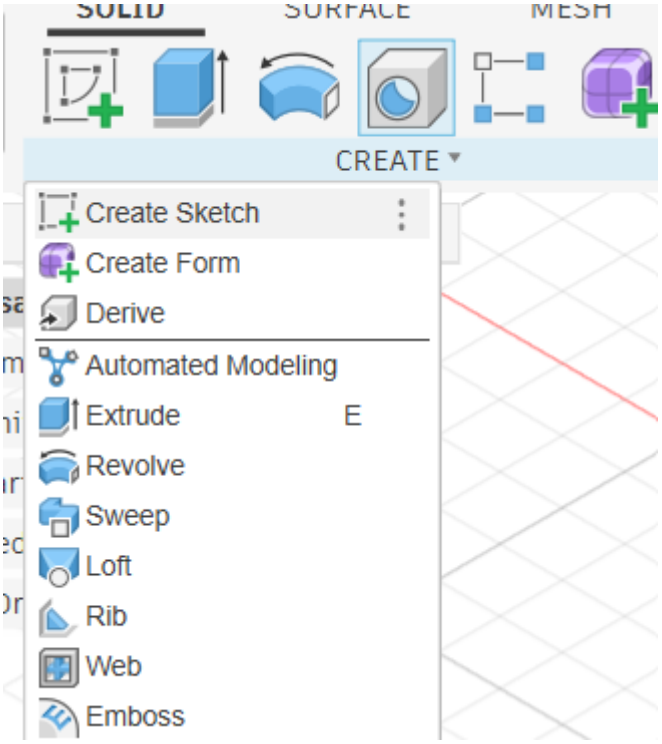
PROCEDURE:

STEP 1: Start a New Design

Open Autodesk Fusion 360 and create a new design. Select the Sketch tool to begin drawing the base profile.

STEP 2: Choose the Working Plane

Select the Top (XY) plane to create the initial 2D sketch.



STEP 3: Draw the Base Profile

Use the Center Rectangle tool to draw a rectangle with dimensions 56 mm × 18 mm. Add the required arc with a radius of 18 mm and a circle with a diameter of 16 mm according to the given drawing. Click Finish Sketch.

STEP 4: Form the Base Solid

Select the required sketch profile and apply the Extrude command. Enter 8 mm as the extrusion distance and confirm to create the base solid.

STEP 5: Sketch on the Top Surface

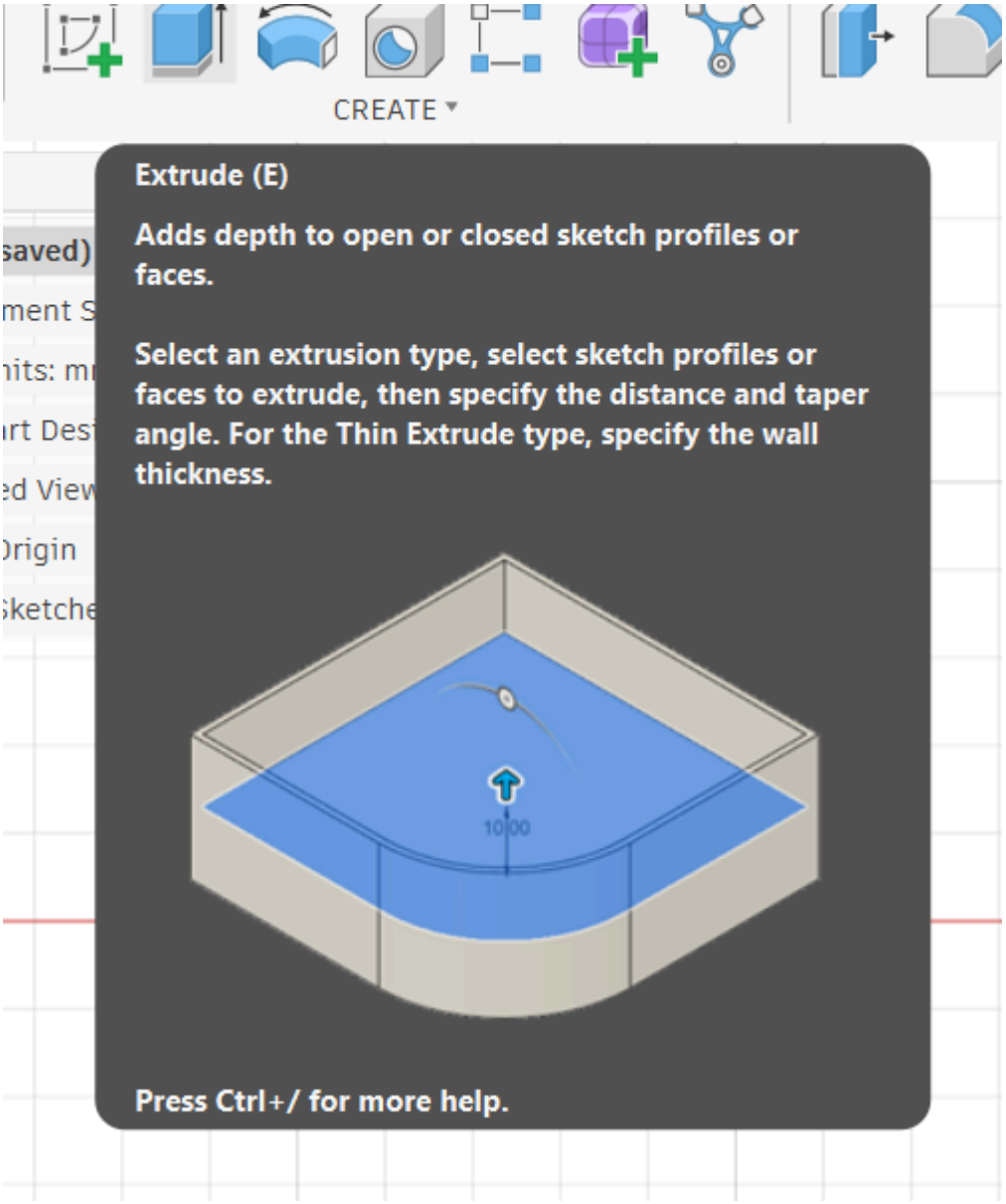
Select the top face of the newly created solid and start another sketch.

STEP 6: Draw the Upper Rectangle

Choose Center Rectangle and create a rectangle measuring 40 mm × 10 mm. Position it according to the isometric drawing and finish the sketch.

STEP 7: Extrude the Rectangle

Select the rectangular profile and extrude it by 32 mm. Confirm the operation to create the raised rectangular feature.



STEP 8: Draw the Triangular Profile

Start a new sketch on the appropriate face. Select the Center Triangle tool and draw a triangle with the specified dimensions of 32 mm × 12 mm. Click Finish Sketch.

STEP 9: Extrude the Triangle

Select the triangular profile and use Extrude. Set the extrusion distance to 6 mm in both directions and confirm the feature.

STEP 10: Create the Polygon Profile

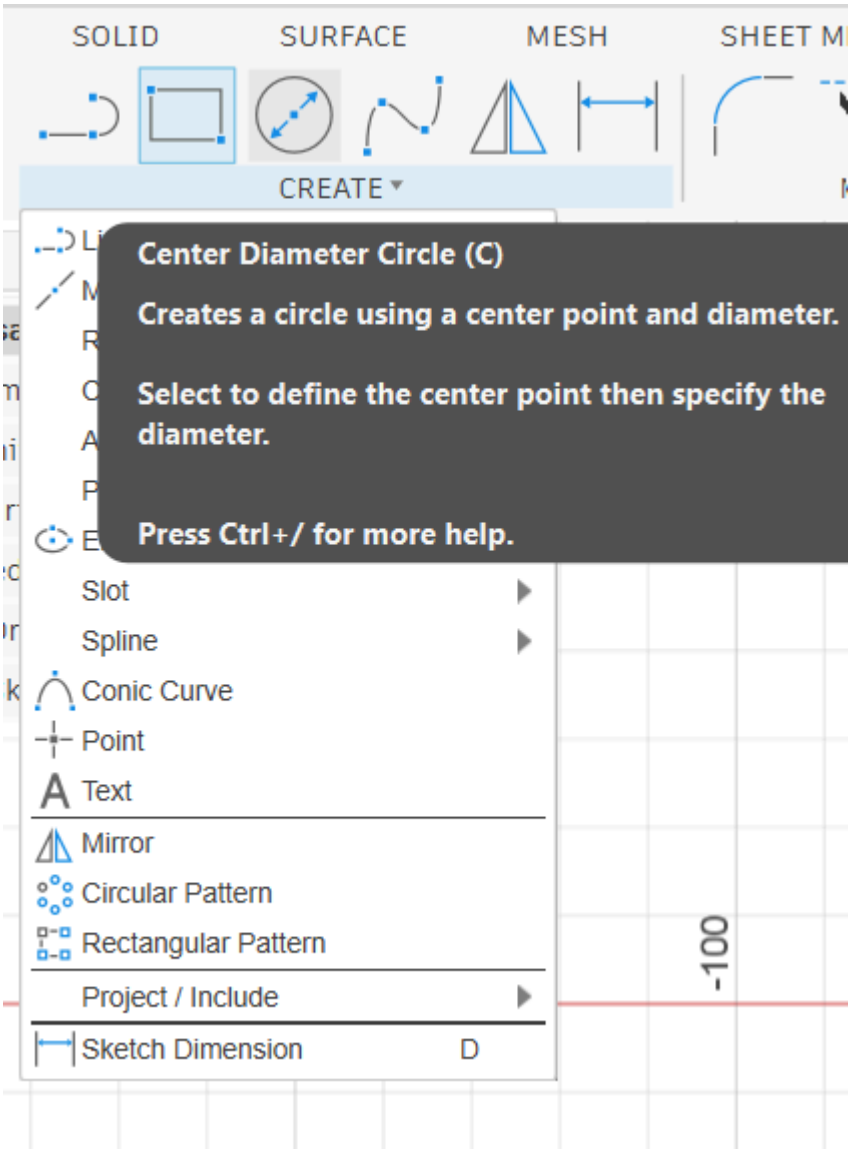
Start a new sketch on the required face and select the Polygon tool. Draw the polygon according to the given dimensions of 40 mm and 30 mm. Set the required number of sides as shown in the drawing and finish the sketch.

STEP 11: Extrude the Polygon

Select the polygon profile and apply an extrusion distance of 10 mm. Confirm to create the polygon-shaped feature.

STEP 12: Draw the Circular Profiles

Create a new sketch on the required end face of the polygon feature. Use the Center Circle tool to draw the required concentric circles with the specified diameters. Finish the sketch.



STEP 13: Complete the Circular Feature

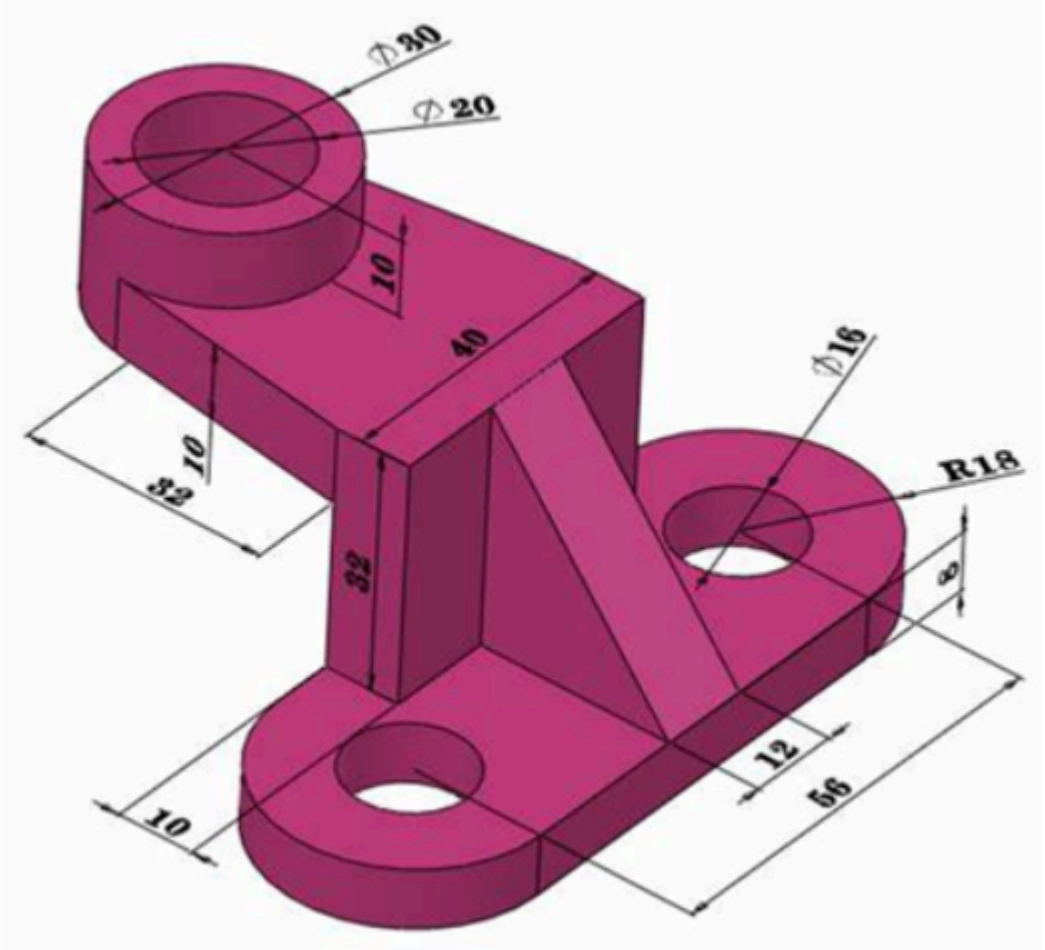
Select the required circular profile and apply the Extrude command with a distance of 10 mm in both directions. Confirm the operation to complete the feature.

STEP 14: Inspect and Save the Model

Rotate the model to inspect it from different angles. Verify the shape and dimensions against the given isometric drawing, make any necessary corrections, and save the completed design.

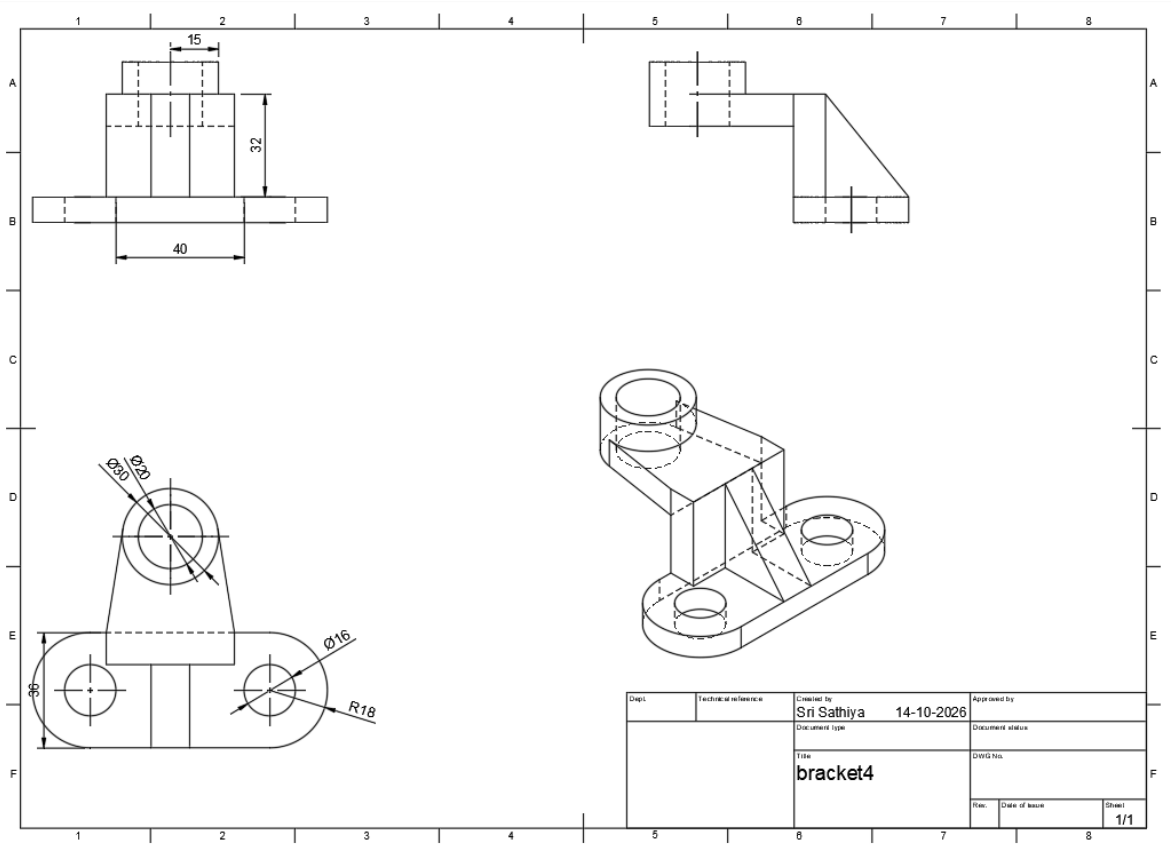
INPUT:

Given isometric drawing.



OUTPUT:

3D solid model created using Autodesk Fusion 360.



***RESULT*:**

Thus, the 3D solid model was created from the given isometric drawing using Autodesk Fusion 360 software.